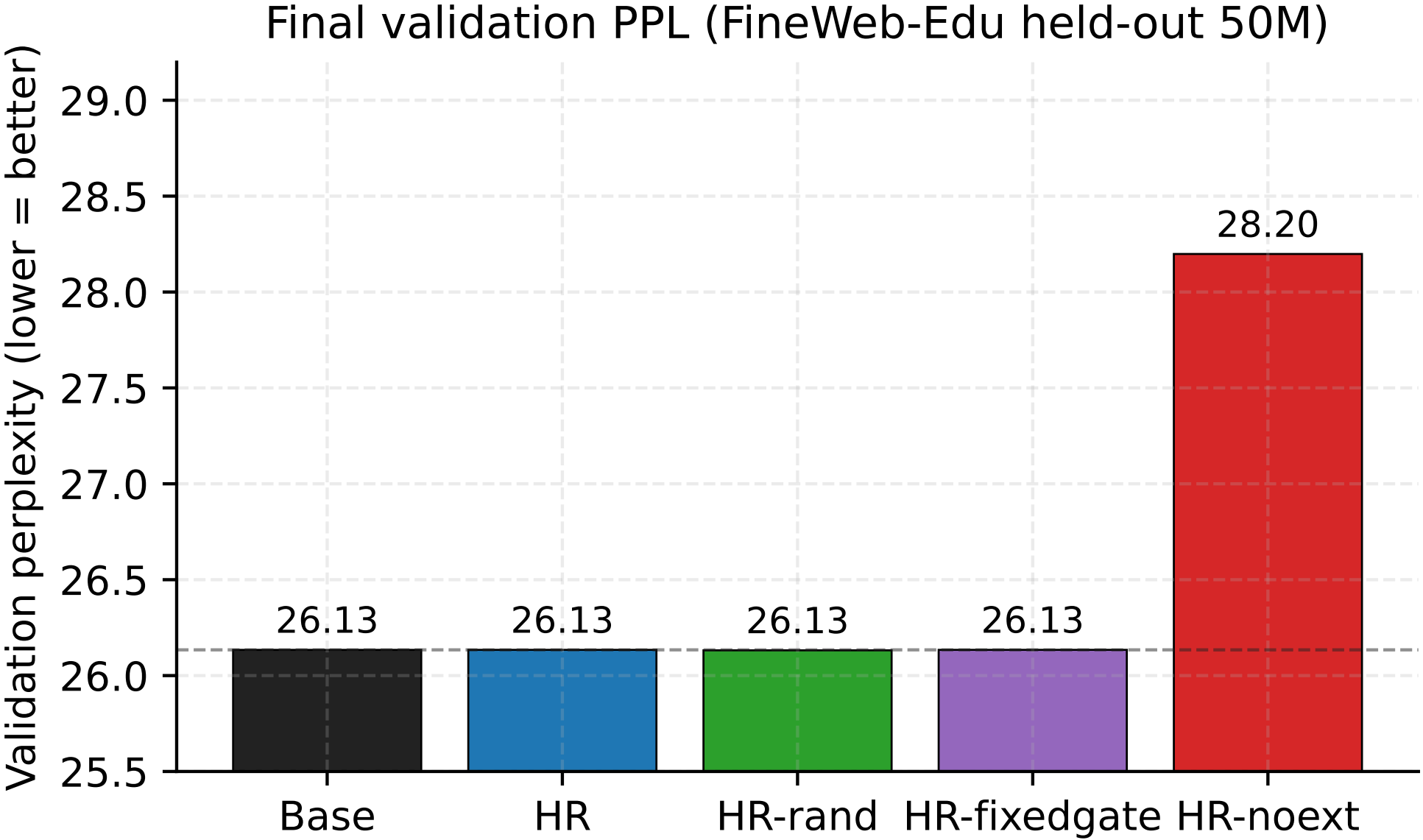
Final validation PPL (FineWeb-Edu held-out 50M)
Validation perplexity (lower = better)
29.0
28.5
28.0
27.5
27.0
26.5
26.0
25.5
26.13
26.13
26.13
26.13
28.20
Base
HR
HR-rand
HR-fixedgate
HR-noext